



CAPSTONE PROJECT **PROPOSAL**

Title: Development of a Web-Based Decision Support System for a Simulated Organization Using Knowledge Graphs and Retrieval-Augmented Generation

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A. General Information

Project Title: Development of a Web-Based Decision Support System for a Simulated Organization Using Knowledge Graphs and Retrieval-Augmented Generation

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B. Executive Summary

Educational institutions increasingly rely on complex strategic and operational decisions made by academic leaders and administrators. Over time, the rationale, context, and outcomes of these decisions are often lost due to staff turnover, fragmented documentation, and dependence on tacit knowledge. This erosion of institutional memory results in inconsistent decision making, repeated errors, and limited organizational learning. Existing decision support systems in higher education largely depend on static reports and structured data, which restricts their ability to capture contextual relationships and support evidence-based reasoning over time.

This capstone project proposes the development of a web-based Decision Support System for a simulated college environment that addresses these limitations through the integration of Knowledge Graphs and Retrieval-Augmented Generation. Organizational entities such as departments, strategies, actions, constraints, and outcomes are represented within a Knowledge Graph to preserve relationships and historical context. Structured decision data is stored in a relational SQL database, while unstructured textual content is encoded as vector embeddings to enable semantic retrieval. A chat interface allows users to interact with the system using natural language, with queries dynamically retrieving relevant information from multiple data sources to produce grounded and context-aware responses.

The resulting prototype will demonstrate how AI-driven decision support can strengthen institutional memory, support learning from prior decisions, and reduce risks associated with knowledge loss in academic organizations.

1. Literature Review

1.1 The Challenge of Organizational Knowledge Loss

Having senior leaders exit organizations means that they leave behind them the knowledge that they gained in decision making. This leaves loopholes that influence future decisions and survival of an organization. This tacit knowledge can hardly be gathered with the help of traditional knowledge management systems (Taherdoost, 2023) (Neiroukh, 2024).

The theory of organizational learning also points out that learning that is stored in routines, processes and personal knowledge forms the basis of an organization in adapting (Miron-Spektor, 2011). The loss of experienced personnel means that organizations lose not only formal knowledge recorded in documents but also contextual information about how and why things were done the way that they were and what forced one to make a move. This loss of knowledge will have a direct effect on the capability of an organization to learn through previous experience and implement the same in addressing future challenges.

1.2 Decision Support Systems in Organizations

The computer relied information systems which can also be called as decision support system(DDS),it uses the information of an organization and compile analyze those information helping in making the decision.It is an amalgamation of data, analyze the whole model as well as interactive interfaces help in making the decision by solving structured and unstructured problems.Use of DSS has help in planning and making the strategy for an organization, performance appraisal and even learning institutions.

1.2.1 Decision Support Systems in Educational Institutions

DDS has increasingly gained a lot of popularity amongs the academic institutions for enabling and helping administrators and other academic schedulers in making decisions on various strategies and effective teaching cycle.The authors of this case study discovered that DDS can also be applied in the context of an educational field whereby the operational and scholarly data is used to make decision about instruction.It showcases that DDS has the capacity of improving the efficiency and accuracy of decision at the institution because of the fact that it clears the need to make judgements based on a subject(Society, 2022)

1.2.2 Big Data–Driven Decision Support Systems

One of the studies published in the Journal of Electrical Systems analyzed the use of the concept of big data analytics, in the decision support system context, as academic data and information continue to swell exponentially.One of the studies proposed the use of a big data-based DSS for education management that would calculate the vast amounts of academic performance data to enable academic plan and prognosis. The findings indicated that DSS using big data and early intervention strategies can be employed to predict academic success. (Systems, 2024)

1.2.3 Importance of DSS in Higher Education

Using Decision Support Systems (DSS) within a management capacity represents an avenue for Higher Education institutions to use information technologies to aid their leaders in integrating the application of information technology into their decision making processes. Research conducted previously indicates that the use of DSS by Administrators will facilitate a more efficient allocation of resources, streamline the Strategic Planning process and provide assistance in meeting student needs through enhanced access to relevant business analytics and actionable feedback. Interestingly, when Higher Education institutions utilize DSS in their academic and administrative functions, they have more efficient operations with less need for manual processing, as well as being able to base their decisions on factual evidence. (Zhang & Goyal, 2024)

1.2.4 Decision Models and Methods in DSS

There have been many studies trying to improve the Decision Support Systems (DSS), through the addition of structured decision-making models. In one study published in 2023, researchers used the Simple Additive Weighting (SAW) technique in the educational version of a DSS that can help solve multiple criteria decision-making (MCDM). This DSS was evaluated by comparing MCDM decisions made using methodologies with DSS against MCDM with no DSS. The study concluded that DSS that include structured decision-making methodologies provide a clearer and more accurate outcome than an MCDM without DSS.(Hak, et al., 2023)

1.2.5 Predictive Decision Support Systems Using Educational Data Mining

There have been many studies trying to improve the Decision Support Systems (DSS), through the addition of structured decision-making models. In one study published in 2023, researchers used the Simple Additive Weighting (SAW) technique in the educational version of a DSS that can help solve multiple criteria decision-making (MCDM). This DSS was evaluated by comparing MCDM decisions made using methodologies with DSS against MCDM with no DSS. The study concluded that DSS that include structured decision-making methodologies provide a clearer and more accurate outcome than an MCDM without DSS.(Saa, 2016)

1.2.6 Data Warehouse and Business Intelligence–Based DSS

One other option for enabling analytics scalability and practical use is data warehousing -based DSS architecture. Created in 2021, it relies on data from independent educational marts. It's a data-centric DSS that leverages business intelligence reporting and trend-analysis tools. This paper has shown that decision support systems (DSS) that are backed by well-designed data warehouses can be trusted to be useful in schools and other learning environments.(Hamoud, et al., 2021)

1.3 Limitations of Traditional Decision Support Systems

The limitations associated with legacy decision support systems are that they lack agility to adjust to dynamic decision environments since they focus heavily on rigid processes like formal reporting methods and rules-based algorithms. They fail to capture and replicate tacit knowledge harvested from historical organizational decisions. As such, traditional DSS can not comprehend natural language and therefore cannot be utilized by non-technical users. Above all, this limitation points to the gap that necessitates continued efforts in intelligent knowledge-driven solutions that aid context-based reasoning and decision-making.

1.4 Knowledge Graphs as Structured Knowledge Representation

Knowledge graphs use semantic technologies and graph theories to represent entities, relations, or attributes among interconnected systems (Gutiérrez, 2021).They rather than rely on flat databases, chart how one concept relates to another to allow for reasoning and inference. This

method has proved effective in sectors such as healthcare and manufacturing where how one entity relates to another is critical in decision-making. (Chandak, 2022) (Bader, 2020)

1.5 Retrieval-Augmented Generation for Grounded Responses

Large language models often generate inaccurate information when they lack domain-specific knowledge. RAG addresses this by retrieving relevant information from external sources before generating responses (Lewis, 2020). This method bases the result on real data as compared to simply relying on the training of the model. Nevertheless, conventional RAG recovers separate text fragments, depending on their similarity, whereas the connection between chunks of information is absent. . (Gao, 2023).

1.6 Combining Knowledge Graphs with RAG

Linking Knowledge Graphs to RAG can create business opportunities. Previous studies show that using knowledge graphs with RAG improves how well information is found and how accurate responses are. For example, the IKEDS framework achieved an accuracy rate of 85.7 percent in financial, health, and supply chain decisions by combining structured knowledge with generative capabilities (Wang, 2025).). Similarly, KG2RAG discovered that using knowledge-graph-based expansion and reorganization led to better retrieval F1 scores, increasing from 0.357 to 0.436 compared to retrieval using semantics (Zhu, 2025). Combining both methods shows connections between retrieved content at the fact level, which makes answers more logical and relevant to the context. These methods are often used for information access in general, and they can also assist with retrieval and providing necessary responses. A lot of time is spent keeping up with and rediscovering organizational decision-making information, where past knowledge, connections, and outcomes are crucial for future decisions.

1.7 Educational & Domain-Specific KG-RAG Systems

The application of KG-RAG in domain-specific systems and instructional systems has also been examined in other recent works. When responding to academic inquiries, the application of KG-RAG and retrieval-augmented generation as well as ordered didactic information known as KA-RAG has shown even stronger reasoning power. The startling accuracy of 91.4 has been demonstrated by KA-RAG. The proficiency of dialogue-based learning systems with KG-RAG has been proven. A systematic linkage of various educational facts has been made possible by the cross-domain expertise in graph design.

1.8 Literature Review Process

A systematic and structured search was performed to locate the body of scholarly work on the topics of decision support systems, knowledge graph, retrieval-augmented generation, and organizational knowledge management. Academic databases such as Google Scholar, IEEE Xplore, ACM Digital Library, Science Direct, and Springer Link were used to find peer-reviewed journal articles, conference proceedings, and technical reports. Retrieval-augmented generation (RAG), educational data mining, decision support systems, loss of organizational knowledge, knowledge graphs, and AI in education management were the search terms.

The studies have been chosen according to their relevance to the field of decision-making support, AI-based knowledge representation, or the use of knowledge graphs or RAG technologies in an organizational or teaching setting. The focus was on the recent materials (2016-2025) to provide up-to-date technological knowledge, and the earlier foundational studies were also provided where needed to provide theoretical background. Articles with unclear methodologies, lack of credibility or directness concerning knowledge retention or decision support were sifted out. Overall, out of the over thirty screened papers, fourteen main studies were chosen to be analyzed and synthesized in this review.

2. Review of Existing Systems

There are some systems that try to mix knowledge graphs with RAG. The GraphRag used in Microsoft utilizes community-based graph summary but makes use of pre-summaries that restrict flexibility. LightRAG is an alternative with light weight, yet is less accurate (F1: 0.293). IKEDS framework offers the best opportunity of total integration that involves five interconnected parts and has been used with good outcomes in a variety of domains. KG2RAG pays particular attention to retrieval enhancement by means of the expansion of the chunk with the use of a graph. A comparison is made of these systems in Table 2. Existing decision support systems mainly rely on static reports and predefined rule-based logic. Such systems work well for structured data but perform poorly when dealing with tacit knowledge and complex organizational relationships. Additionally, traditional DSS do not support natural language querying, making them less user-friendly for non-technical users. These limitations highlight the need for intelligent, AI-driven decision support approaches.

Table 2: Comparison of existing KG-RAG systems.

System	Strengths	Limitations
GraphRAG	Global context awareness, query-focused summarization	High computational cost, pre-computed summaries limit adaptability

LightRAG	Fast processing, lower resource requirements	Lower accuracy (F1: 0.293), limited reasoning depth
IKEDS	85.7% accuracy, cross-domain reasoning, explainability	Complex architecture, high knowledge engineering effort
KG2RAG	Improved retrieval (F1: 0.436), coherent context	Focused on QA tasks, limited decision support features

3. Research Gap

In addition, most of the existing studies evaluate their methods on a benchmark/real company basis, and there is minimal research on simulated organizational settings to validate their systems and to experiment. Although current systems have encouraging scores in the field of question-answering exercises, they have not explicitly focused on the knowledge retention to decision making in an organization. Recent methods are missing: (Taherdoost, 2023) purpose-built ontologies to represent structures of organizations such as departments, strategies, and other outcomes; (Neiroukh, 2024) user-friendly web interfaces on querying past decisions; and (Gutiérrez, 2021) the easily accessible web interfaces. This project will seal these gaps by coming up with a web-based decision support system that will suit the organization's knowledge management.

Based on the literature reviewed, traditional decision support systems fall short in capturing tacit organizational knowledge, while new techniques such as knowledge graphs and retrieval-augmented generation bear a high potential for improving contextual reasoning and information retrieval. However, the current applications are mostly intended for question-answer or general information retrieval purposes, not for maintaining the history of organizational decisions to be used later for strategic decisions.

The proposed project is intended to bridge this knowledge gap and create a web-based intelligent decision support system that integrates knowledge graph representation and retrieval-augmented generation methods. It will store the organization's past decision-making, situational connections, and outcomes in a structured graph format and grant natural language querying through an interface utilizing artificial intelligence. By integrating structured organizational knowledge with the generative functions of artificial intelligence, this project hopes to enhance organizational knowledge retention, improve evidence-based decision making, and enable friendly analytical tools for the primarily academic or organizational management institutional clients.

C. Project Purpose

Problem Statement

Colleges and other educational institutions often experience challenges in the reuse and preservation of strategic and operational decision-making knowledge. Information about past decisions made by the academic and administrative staff is mostly fragmented across different sources, including reports, policies, emails, and the knowledge held by the staff. This has made it difficult for these institutions to retain knowledge about past decisions, which has resulted in inconsistencies in the decision-making process. This has often come at a price, especially when the key staff members depart the institution. This is often referred to as corporate amnesia. (Wikipedia, 2024)

Evidence 1: 42% of Institutional Knowledge Is Unique to Each Individual

Fact: When an employee quits their peers cannot perform 42 percent of their job due to the fact that such a person has a unique knowledge (Panopto, 2018) that he or she does not share with anybody.

Source: surveying over 1,000 workers in the U.S. the first study of its kind to examine workplace knowledge and productivity reveals that half of the knowledge in institutions is individualized and that 12 to 15 years on average, one large enterprise is wasting \$47 million in productivity due to poor knowledge sharing (Panopto (n.d.), Knowledge and Productivity Workplace Report).

This has often come at a price, especially when the key staff members depart the institution. (Galdamez, 2025)

Evidence 2: Peer-Reviewed: 91-Study Systematic Review of Knowledge Loss from Turnover

Fact: Past academic literature ascertains that the turnover of organizational staff members leads to quantifiable loss of routines, memory and absorptive ability.

Source: A two-part synthesis of systematic literature published in The Learning Organization (Emerald Publishing) based on 91 empirical studies on knowledge loss induced by turnover found that knowledge loss through turnover results in a wide range of organizational-level effects, such as the erosion of vital routines, organizational memory loss, and reduced absorptive capacity. (Galan, 2023) (Galan, Knowledge loss induced by organizational member turnover: A review of empirical literature, synthesis and future research directions (Part II), 2023)

The current state of decision support systems in colleges is mostly based on reports or isolated information systems, which are not aware of the historical context and cannot integrate structured data with unstructured knowledge. These systems are not capable of providing the necessary depth of analysis required for effective decision-making. This has made it difficult for these institutions to learn from past decisions, which is critical in the decision-making process. Without the use of a centralized and intelligent system for the preservation and organization of

knowledge, institutions can be vulnerable to repeated errors, lost chances and opportunities and overall reduced effectiveness.

Project Objective:

The project seeks to develop a centralized decision support system that can be accessed via the internet for a simulated college, addressing the aforementioned challenges. In particular, the objectives are:

1. Design a web-based decision support system that can capture and retain decision-making knowledge for various departments.
2. Implement a SQL database for the decision-making knowledge for strategic and operational decisions.
3. Implement a Knowledge Graph for modeling organizational entities, such as strategies, actions, constraints, and outcomes.
4. Implement the Retrieval-Augmented Generation model with the Knowledge Graph for precise, contextual, and transparent decision-making.
5. Implement the capability for users to query historical decision-making via natural language, retrieving relevant historical cases.
6. Analyze historical decision outcomes for successful strategies and unsuccessful strategies.
7. Demonstrate the potential for decision support systems based on artificial intelligence for organizational learning, continuity, and risk minimization.

No.	<i>Proposed Functionality</i>	<i>Problems Solved/ Opportunities</i>
1	Centralized Knowledge repository	Solves the issue of scattered information on decisions made, thus avoiding the loss of institutional memory.
2	Knowledge Graph-Based Modeling	

		Provides a structured way of representing relationships within organizations, thus facilitating relationship-based reasoning.
3	Natural language query interface	Solves the issue of accessibility of historical data to non-technical users.
4	Retrieval-Augmented Generation (RAG)	Guarantees that the decisions made are based on facts and are context-sensitive, thus avoiding the issue of decisions based on assumptions.
5	Decision outcome analysis	Allows learning from past experiences of success or failure, thus directly addressing inconsistencies in decisions.
6	Flexible input interface (text and PDF upload)	Allows real-world context to be integrated into the system, thus avoiding the limitation of using structured or isolated data.

D. Assumptions

The project is based on the assumption that the organizational system, the system of governance, and the decision-making system of an actual school have been accurately replicated in the simulated college environment. In order to carry out meaningful analysis and offer effective decision support, it is assumed that the simulated historical decisions data is exhaustive, consistent, and representative of realistic issues. It is further assumed that the target group, including administrative and academic staff, is aware of web technology and can use the natural language system without special training.

From the technical point of view, it is expected that the relational database, the knowledge graph, and the vector embedding pipeline have been designed properly to represent the data and relationships of an institution in an effective way. The large language model is expected to produce answers that are purely based on retrieved knowledge of an organization, and hallucinations should be minimized with controllable retrieval. The system is not expected to be used as a decision-making tool but as a decision support tool, and actual decision-making is expected to be carried out by the actual stakeholders of an institution.

E. Project Description, Scope and Management Milestones

1. Project Description

The project requires designing and developing a web-based, centralized, and decision support system for a simulated college setting. The simulated college setting comprises Operational Academic Units and Parent Regulatory and Administrative Departments. The system aims at capturing, storing, and reusing knowledge for decision-making in the academic and administrative fields for the purpose of decision-making in the long run. The solution that has been proposed for the project is a knowledge management solution that utilizes a hybrid model for knowledge management, which combines Knowledge Graph, structured databases, and Retrieval Augmented Generation.

The process starts with the inputting of the decision-related inputs in the form of the decision problem, which means it is in the context of the constraint or the supporting document. The inputs that have already been fed into the system are encoded and stored in different data stores. These data stores include the structured SQL data store, the graph data store for the semantic relationships, and the text embeddings vector data store. The process of decision-making involves the processing of the user queries. This is done by the large language model, which refines the queries and forwards them to the relevant data streams. The knowledge retrieved is then unwound and fed back into the large language model for the creation of grounded and context-sensitive responses.

The target users of the system are the administrators of the college, the department heads, and the academic planners. The users are benefited by the fact that the system allows them access to the decisions made in the past by the university in natural language.

2. Project Scope

Platform and Users

The project will provide a web-based platform which will be located in a simulated college. The simulated college will consist of **Operational Academic Units** such as Business Administration Department, Information Technology Department, Management Department, Social Sciences Department, Hospitality Department, and **Parent Regulatory and Administrative Department** which includes Finance Department, Human Resource Department, and Quality Assurance and Accreditation Office. The users of the project will be academic administrators who can interact with the system via a web browser interface.

Technology Stack

The application utilizes a state-of-the-art, high-performance stack, with the frontend implemented with React, TypeScript, and the Vite framework, along with Zustand for state management, TanStack Query for data fetching and caching, and Tailwind CSS for rapid UI development, while

the backend has been implemented with Python, FastAPI, and the Pytest testing framework. The AI/RAG layer utilizes LangGraph/LangChain for the orchestration of document chunking, retrieval, and synthesis with an open-source model for cost-effective synthesis, along with pgvector for embeddings and a polyglot persistence layer utilizing PostgreSQL, Neo4j, and pgvector. The infrastructure stack utilizes a zero-cost model, with the frontend deployed on the Vercel platform, the backend deployed on the Render platform, and the databases deployed on Neon and Neo4j Aura.

Client-Side Functionality

The client side has input forms for decision context, which are arranged with the aim of uploading the supporting materials, thus ensuring good and accurate data acquisition. The client side has a natural language query, which allows the user interface with the decision-making system with the natural language, which does not require any technical skill. The client side displays the recommendations of the decision, as well as the cases in the history, with the aim of making an informed decision, thus enhancing transparency and enabling the user to know how the conclusions can be made with the knowledge.

Server-Side Functionality

The API services for the ingestion of the data and the retrieval of the same are high-performance asynchronous API services built with FastAPI. It supports integration and testing with API documentation. A hybrid approach is used for the storage of the data to utilize the capabilities of the systems. For example, the structured data of the decision is stored by PostgreSQL, the knowledge graph is stored by Neo4j, and the vector store is used to determine the semantic similarities. It was decided to use this method rather than the database solution to achieve the best performance. The selection of the applicable data sources and the reason paths is achieved by the application of the LLM-based routing and retrieval pipelines to the quality of the decision support. The knowledge graph constructions and queries are used to determine the insights based on relationships that are hard to determine by relational databases. The workflows of LangGraph are used to control the state and the logic of the reasoning process.

Project Accomplishments

The project will illustrate how a hybrid AI-driven system can improve decision continuity, institutional learning, and explainability in a college setting.

Boundaries and Limitations

The system is expected to function in a simulated college environment to show the viability of the suggested approach to decision support. Real-time integration of live systems of institutions, e.g., student information systems, etc., is out of the scope of the project. The system is expected

to aid in decision support and suggestions, not the actual execution of the decision, to ensure human intervention in decision-making. Performance evaluation of RAG will be carried out by Precision and Recall at K, to make contextually relevant decisions, with the aid of a synthetic evaluation dataset. The evaluation dataset will consist of what text or what title of the text is, etc., and will be compared to evaluate the output. For vector embeddings, the initial data will consist of policies of all the departments in the college, and then the data will increase as more data is fed into the system by the users, stored in KGs, relational databases, and embeddings, etc. Every step of the system will be recorded to make it observable and easier to debug.

3. System Flow Diagram

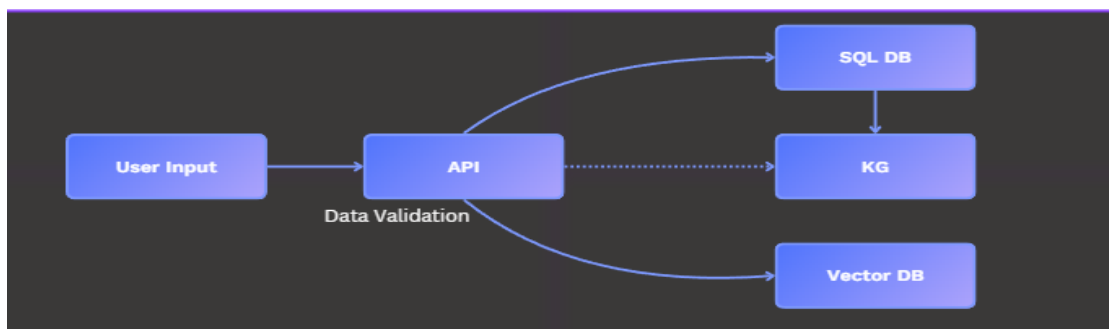


Figure 1 System flow for Knowledge Base creation

In this diagram, there is a data ingestion pipeline in which data from User Input flows through the API, resulting in Data Validation, after which data is stored in a polyglot architecture for data persistence. In this architecture, data is stored in three different data storage systems, i.e., a SQL DB, a Vector DB, and a KG, each of which stores a specific type of data, thus optimizing the storage of data for future querying and analysis.

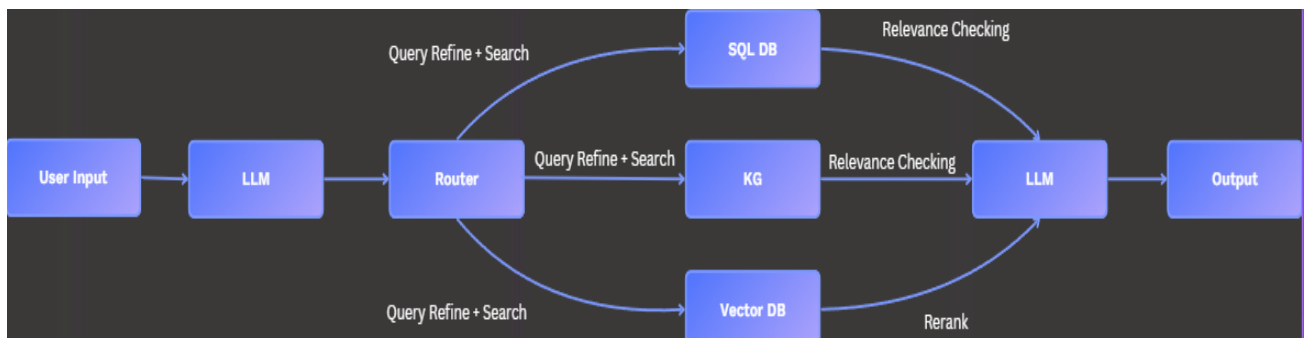


Figure 2 : System flow for LLM Query

The above diagram depicts a Retrieval Augmented Generation (RAG) model, which can deal with complex queries. It begins with the processing of the User Input by the first LLM, which refines the input query. This is followed by the input query going through a Router. This router routes the input query efficiently to the best possible data source, which could be a SQL DB, a Knowledge Graph (KG), or a Vector DB, based on the input query. When the information is retrieved after going through the process of Relevance Checking or Reranking, the final LLM produces the Output.

4. Summary of Milestones and Deliverables

The project will be developed using an iterative and modular approach. Key milestones and deliverables are outlined below.

Milestone No.	Milestone	Deliverables	Start Date	End Date	Duration	Person Responsible
1	Group Formation and Topic Finalization	Finalized team structure and approved project topic	7 Jan 2026	30 Jan 2026	24 days	Samrat
2	Initial Proposal Defense	Proposal presentation and feedback	1 Feb 2026	3 Feb 2026	3 days	ALL
3	Proposal Submission	Approved written project proposal	4 Feb 2026	10 Feb 2026	7 days	ALL
4	Requirement Analysis and Detailed System Design	Functional requirements, system architecture, data flow diagrams	11 Feb 2026	24 Feb 2026	14 days	Samrat

5	Database Schema and Knowledge Graph Design	SQL schema, KG ontology, entity and relationship definitions	25 Feb 2026	10 Mar 2026	14 days	Rayan, Sanjina, Samrat
6	Frontend UI Wireframes and Initial Setup	UI wireframes, React project setup	11 Mar 2026	24 Mar 2026	14 days	Ruby, Bibisha
7	Backend API Design and Setup	FastAPI project structure, initial endpoints	25 Mar 2026	7 Apr 2026	14 days	Samrat, Rayan
8	Capstone 1 Review and Documentation	Design report, interim demo	8 Apr 2026	14 Apr 2026	7 days	ALL
9	Knowledge Graph Implementation	Neo4j graph populated with simulated data	15 Apr 2026	29 Apr 2026	15 days	Sanjina, Samrat
10	RAG and LLM Pipeline Integration	Retrieval pipelines, grounding logic, query routing	30 Apr 2026	20 May 2026	21 days	Samrat
11	Frontend Feature Completion	Query interface, recommendation display, document upload	21 May 2026	10 Jun 2026	21 days	Ruby, Bibisha

12	System Integration and Dev-Ops	End-to-end system integration	11 Jun 2026	25 Jun 2026	15 days	Samrat, Rayan
13	Testing and Performance Evaluation	Functional tests, simulated scenario evaluation	26 Jun 2026	10 Jul 2026	15 days	Sanjina, Rayan
14	Final Documentation and Report Writing	Final project report and user documentation	11 Jul 2026	5 Aug 2026	26 days	ALL
15	Final Presentation and Submission	Demo, presentation slides, final submission	6 Aug 2026	21 Aug 2026	16 days	ALL

Gantt Chart:

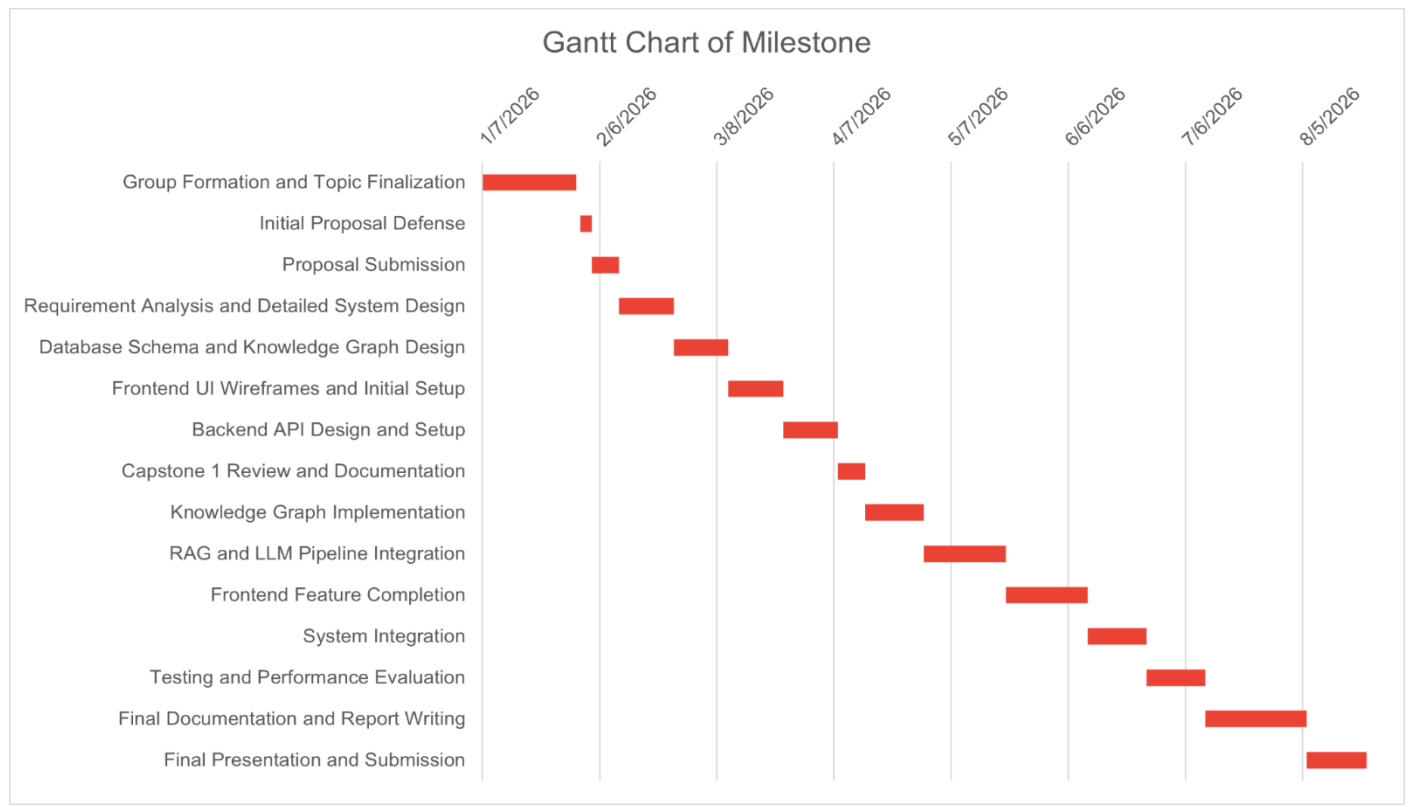


Figure 3 : Gantt Chart of Milestone and Deliverables

The Gantt chart depicts the entire activity of the project from January 2026 to August 2026, including the order of activities, time period, and overlapping of the key activities of Capstone 1 and Capstone 2. From the figure, it is clear that there is a necessity for the first stage of the project, which is aimed at the project start-up activities such as group development, topic narrowing, proposal defense, and submission. The activities of this stage provide the base of the project.

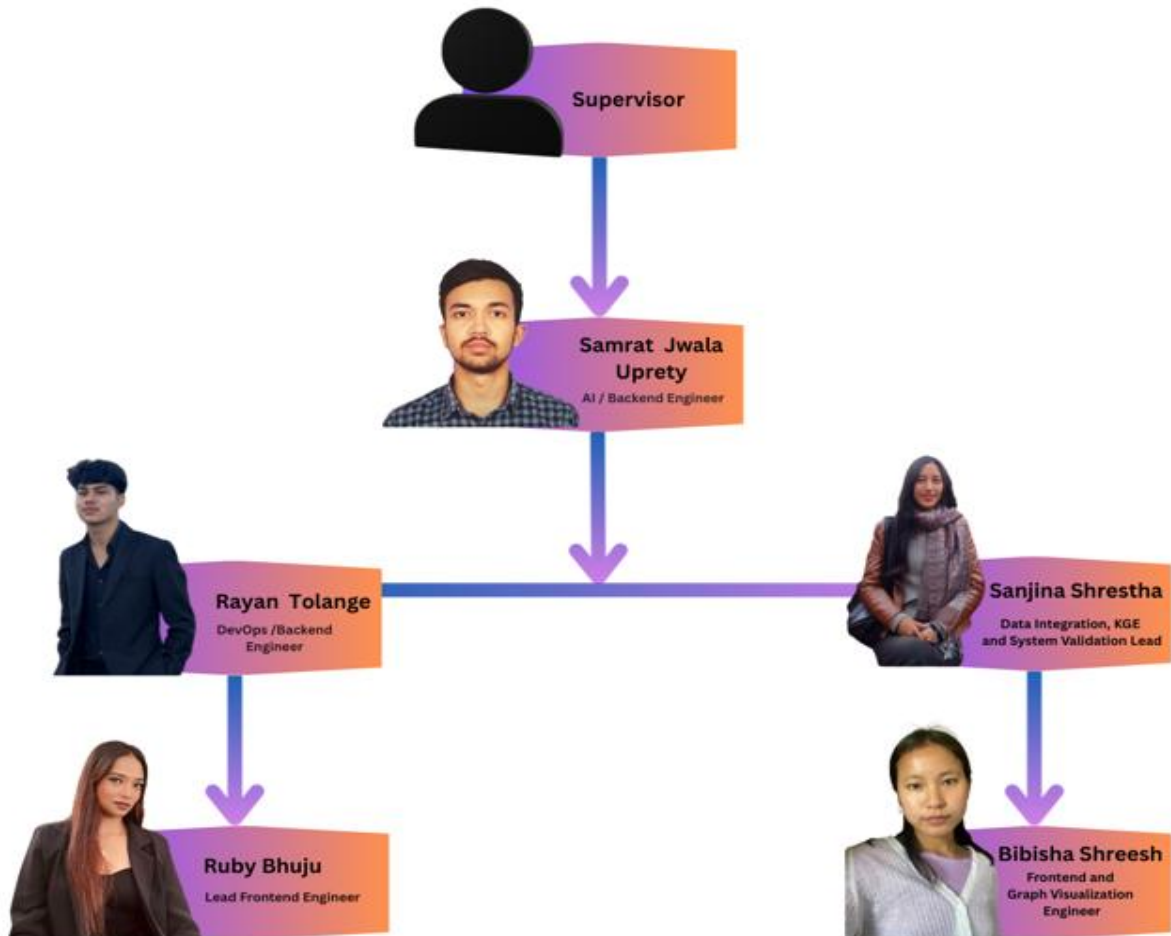
The milestones of Capstone 1 include requirement analysis, system architecture design, knowledge graph modeling, database schema development, and the first frontend and backend. This stage is aimed at the idea design and implementation to ensure all the technical aspects are correct before the development on a large scale can take place.

The process of implementing, integrating, and evaluating the project is referred to as the capstone 2 process. The major activities in this process include the implementation of the knowledge graph, the implementation of the retrieval augmented generation and the integration of the large language model, the implementation of the frontend features, the integration of the systems, and the performance test of the system based on the simulated data sets provided. The final milestone activities include documentation, reporting, and presentation of the project. In

summary, the Gantt chart will illustrate an outline of the systematized stepwise procedure, which is vital for the timeliness of the delivery process.

F. Project Organization:

1. Project Organization Chart:



2. Organization Description:

The project team is established to deal with all the technical requirement of the system and ensure that there is a definite chain of command.

Composition: To address the technical specifications of the project, the team of five people was structured based on their experience in areas of Artificial Intelligence, Graph Database Theory, Software Engineering and User Experience Design.

Power and Control: The project supervisor is the final authority regarding the academic requirements and approval of the project scope. The Lead Person is an operational leader capable of delegating tasks, developing a schedule and acting as a channel of communication between the technical team and the stakeholders.

Communication Plan: A well-organized communication system is followed by the team to make sure that effective collaboration is well managed. This will consist of asynchronous updates in the form of version control systems (GitHub) daily, asynchronous sprint planning in the form of Google meets and WhatsApp to arrange on a daily basis and provide short-updates weekly, and monthly progress meetings with the Project Supervisor.

3. Roles and Responsibilities:

Full Name	Roles	Responsibilities
Samrat Jwala Uprety	Project Lead, AI Architect and Backend Engineer	Builds the main AI logic to make the system give useful and intelligent responses. Develops the primary communication channel between the user and the AI. Monitors the general technical flow, and application cleverness.
Rayan Tolange	Infrastructure and DevOps Engineer	Maintains databases containing all organization information which are organized. Make sure the system is stable, secure and running without failure. Manages the technical infrastructure to host, operate and upgrade the application.
Ruby Bhujju	Lead Frontend Engineer	Constructs the site design, buttons and user dashboard. Make sure that the system is simple to use and navigate by non-technical managers. Controls the method in which the users enter their questions and the way they see the answers of the AI.

Bibisha Shreesh	Frontend and Graph Visualization Engineer	Prepares visual maps and charts indicating the relationships between various decisions. Facilitates the construction of the web site so that information that is complicated can be presented openly. Aids in the process of searching and retrieving information within the system to present itself to the user.
Sanjina Shrestha	Data Integration, KGE and System Validation Lead	Gathers, distributes, and organizes the information upon which the system is informed. Maintains the knowledge maps to keep the facts within the system correct. Makes end checks to detect mistakes and ensures that the system is functioning properly.

G. Risk Analysis

In this section, the major risks to the development, integration, and deployment of the proposed Web-Based Decision Support System (DSS) in consideration of the Knowledge Graphs and Retrieval-Augmented Generation (RAG) are outlined. In consideration of the fact that the DSS will be dependent on knowledge representation, artificial intelligence retrieval, and the generation of large language models, there are likely to be a number of risks.

These risks are measured in terms of probability and impact, and suggestions are made to mitigate the risks of incorrect decision recommendations, integration, and delays in project delivery.

1. Potential Risks

R1 – Knowledge Graph Data Quality and Accuracy (Data Risk)

The achievement of the DSS depends entirely on the accuracy and comprehensiveness of the data used to create the knowledge graph.

R2 – Knowledge Graph and RAG Integration Complexity (Technical Risk)

The integration of query systems with other systems may present risks of inefficient query systems and inconsistent data retrieval.

R3 – Model Hallucination and Unsupported Generated Responses (Trust Risk)

The responses generated by the Large Language Model used by the RAG may not be supported by the data retrieved from the knowledge graph.

R4 – Query Performance and Scalability Limitations (Performance Risk)

The performance of the knowledge graph may slow down with an increase in the size of the graph.

R5 – Inadequate Technical Expertise or Tool Limitations (Operational Risk)

The technical team may face challenges due to inadequate knowledge of knowledge graph technology and the RAG framework used by the AI deployment environment.

R6 – Data Retrieval Inconsistency and Context Loss (Quality Risk)

There is a risk of data retrieval being inconsistent, and the DSS may not be able to fetch the relevant context, resulting in a response that may be incomplete or irrelevant to the data entered in the DSS.

R7 – System Integration Failure Between Frontend, Backend, and AI Services (System Risk)

There is a risk of failure in the integration of the user interface, backend APIs, graph databases, and AI services, resulting in a failure of the system itself.

R8 – Timeline Overrun and Project Scope Expansion (Project Delivery Risk)

There is a risk of project delays or scope expansion because of the integration of multiple sophisticated technologies in the project.

2. Risk Assessment Methodology

The risks are assessed based on the following metrics:

Likelihood (L)

1 = Rare

2 = Unlikely

3 = Possible

4 = Likely

5 = Almost Certain

Impact (I)

1 = Negligible

2 = Minor

3 = Moderate

4 = Major

5 = Critical

Risk Score

$R = L * I$ (Range 1–25)

Risk Level Classification

Risk Score	Level	Interpretation	Required Response
1–4	Low	Acceptable Risk	Record and periodically review
5–9	Medium	Manageable Risk	Apply Mitigation If Feasible
10–16	High	Serious Risk	Apply Mitigation Required and Continuously Monitor

Risk Evaluation Table

ID	Category	Risk	L	I	R	Level
R1	Data	Knowledge Graph Data Inaccuracy	4	4	16	High
R2	Technical	KG-RAG Pipeline Integration Complexity	3	3	9	Medium
R3	Trust	Model Hallucination or Unsupported Response	2	2	4	Low

R4	Performance	Slower Graph Queries and Scalability Issues	3	3	9	Medium
R5	Operational	Lack of Expertise or Tool Limitations	2	1	2	Low
R6	Quality	Inconsistency in Retrieval and Loss of Context	2	2	4	Low
R7	System	Integration Failure Between Frontend, Backend, and AI	2	2	4	Low
R8	Delivery	Excessive Timeline and Scope Growth	1	2	2	Low

The following sections outline the mitigation strategies for the project, grouped under Preventive, Detective, and Corrective measures to ensure a robust and reliable system:

Technical & Data Integrity (R1, R2, & R3)

To ensure data integrity in the Knowledge Graph (KG), preventive measures are in place to use curated data and perform data normalization. For detectively addressing the data, the system will require human verification of the relationships between entities and perform periodic checks to maintain consistency in the data.

In terms of the integration complexity of KG-RAG, the strategy proposes a modular architecture and standard APIs to avoid any bottlenecks in integration. Continuous testing of the graph queries and retrieval modules is carried out to monitor integration, and corrective actions of refactoring and fallback are undertaken to ensure the system is operational in the event of integration failures. In terms of the mitigation of Model Hallucination, the strategy proposes limiting the generated output to the retrieved knowledge, and the use of advanced prompt engineering techniques is undertaken to improve accuracy. Accuracy is detected by cross-checking the output with the source, and corrections are undertaken through the use of confidence indicators and mandatory citations.

Performance, Scalability, & Retrieval (R4, R6, & R7)

In terms of Performance and Scalability, the scope of the system is initially limited to simulated data, and optimization of indexing and queries is undertaken to ensure Performance and Scalability. Performance benchmarking and load testing are undertaken as the primary detective tools, and caching and database query optimization are undertaken as corrective actions in the event of any latencies in Performance and Scalability.

Retrieval Inconsistency is averted by improving the quality of embeddings and utilizing a hybrid search strategy. Retrieval Inconsistency is evaluated by executing benchmark queries, which enables making corrective changes to ranking strategies and data preprocessing techniques. To prevent System Integration Failures, API communication standards and services are defined. Continuous system testing helps detect silos and failures, which are fixed by strengthening error logging and fallback services.

Resource & Project Management (R5 & R8)

To address Limited Technical Expertise, open-source tools are utilized with good documentation and adequate time provided for experimentation by the team. Periodic technical reviews and prototyping help detect technical knowledge gaps. As a corrective measure for Limited Technical Expertise when technical issues are insurmountable, open-source resources are utilized and the tool stack is simplified.

To prevent Timeline Overruns, the project is implemented by first creating an MVP with defined project milestones. Progress is monitored weekly by performing weekly reviews to detect delays. As a corrective measure for Timeline Overruns, the project reallocation of resources for development is performed to prevent delays in the delivery of the main system.

H. Appendices

Reviewed Literature	Concept/Idea	How are we different / How can we add value
Taherdoost & Madanchian (2023) – AI and Knowledge Management: A Partnership	Research the application of Artificial Intelligence in knowledge management systems. Demonstrates ways in which AI can make knowledge retrieval, sharing and decision-making more efficient and points to organizational issues in managing large volumes of knowledge.	Specifically, our project will involve the incorporation of RAG and Knowledge Graphs into a Decision Support System to apply AI-based knowledge management. As opposed to general knowledge management discussions, our system employs structured organizational knowledge and semantic relationship to offer real time explainable decision support.
Neiroukh et al. (2024) – AI Capability and Organizational Performance	Researches the role of AI capabilities in enhancing decision-making, operational performance and knowledge	Our system takes these theoretical AI abilities and transforms them into a practical web-based DSS prototype. It integrates generative AI models with knowledge graphs to

	discovery to enhance organizational performance.	come up with valuable recommendations that are aligned with specific instances of organizational decisions.
Gutiérrez & Sequeda (2021) – Knowledge Graphs	Defines knowledge graphs as semantic graph-based models that enhance the connectivity of the information and the contextual representation of knowledge by modeling entities and relationships.	Our project adopts the knowledge graphs as the primary data modeling layer in order to represent organizational procedures, policies, and situations of decisions. This can be used to retrieve relationships and increases explainability of decisions.
Chandak et al. (2022) – Building a Knowledge Graph for Precision Medicine	Shows how knowledge graphs could be employed to aid clinical decision-making in healthcare through the combination of heterogeneous medical data.	Though the focus of this paper was on healthcare applications, our project shows that the Knowledge Graph methodology is not only limited to domain-specific medical applications but can also be applied to organizational decision environments.
Bader et al. (2020) – A Knowledge Graph for Industry 4.0	Describes how knowledge graphs are being applied in manufacturing to assist intelligent industrial decision systems and combine production data.	Our system extends the application of Knowledge Graphs outside of knowledge intensive organizational decision support to industrial automation. To be able to synthesize information and generate recommendations, we integrate KG with RAG-based generative AI.
Lewis et al. (2020) – Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Introduces the RAG architecture that integrates both text generation and information retrieval to improve contextual response as well as factual accuracy of NLP systems.	Knowledge Graph-guided context enrichment and integration of structured organizational data are used in our project to enhance the quality of retrieval and to keep RAG as the main reasoning mechanism.
Gao et al. (2023) – Retrieval-Augmented	Gives a detailed overview of RAG architectures and outlines the shortcomings, including the ability to retrieve	Our system tries to resolve those limitations by applying Knowledge Graph-based retrieval that enhances the contextual understanding,

Generation for LLMs: A Survey	single document fragments without recording the semantic connections between them.	relationship awareness, and multi-hop reasoning.
Wang et al. (2025) – Construction of Intelligent Decision Support Systems (IKEDS)	Introduces a smart DSS architecture that combines AI techniques with knowledge integration and proves to be very accurate across a variety of fields.	This framework is extended into our project where modern LLM-based RAG architecture and Knowledge Graphs are incorporated to increase the explainability, scalability, and domain adaptability in a simulated organizational environment.
Zhu et al. (2025) – Knowledge Graph-Guided Retrieval Augmented Generation (KG2RAG)	Proposes KG-directed chunk growth procedures to enhance the accuracy of retrieval and the depth of reasoning, with better F1 scores in benchmark QA datasets.	Our system uses KG-RAG integration in a similar way but in decision support cases rather than QA tasks and does not simply answer questions but produces actionable recommendations.
Li et al. (2025) – KA-RAG: Integrating Knowledge Graphs and Agentic Retrieval-Augmented Generation for Intelligent Educational QA	Introduces the idea of an educational QA system combining Knowledge Graphs with agentic retrieval system, which can enhance a higher retrieval accuracy and domain-specific question answering.	KG-RAG methodology is applied to organizational DSS (and not educational QA) in our project. We concentrate on strategic decision support, domain modeled knowledge and recommendation generation that can be interacted with by the user.
Zhang et al. (2025) – Generating Vicarious Dialogue for Online Learning Using KG-RAG	Shows application of KG-enhanced RAG to create interactive learning dialogue and creation of contextual content.	KG-RAG is used in our system not in the educational dialogue but in the production of structured decision explanations and recommendations in accordance with the organizational workflow.
Nguyen et al. (2024) – Cross-Data Knowledge Graph Construction for Educational QA Systems	Presents a system that uses multi-source data integration to construct knowledge graphs in domains to provide educational answers to questions.	Our system expands the cross-data KG construction into simulated organizational datasets, combining policies, operational data, and external knowledge sources to support decision making.
Zhang & Goyal (2024) – AI-Driven Decision	Illustrates the importance of AI-based DSS in enhancing	Our project extends DSS implementation to non-higher

Support System Innovations to Empower Higher Education Administration	strategic planning and resource allocation, as well as the administrative decision-making process in higher educational institutions.	education sectors through a simulated organizational setting, including Knowledge Graphs and RAG to enhance the general decision-support.
Abu Saa (2016) – Educational Data Mining & Students’ Performance Prediction	Shows how to use data mining methods in predicting student performance and enabling the educational decision-making process.	Our project builds upon the predictive analytics by providing the concepts of decision support on top of knowledge-based reasoning with AI, Knowledge Graphs, and RAG, and can offer explainable and context-aware decisions.

References

- Anon., 2022. Educational data mining: prediction of students’ academic performance using machine learning algorithms. *Smart learning environment*, 9(11).
- Anon., 2025. *The Importance and Applications of Decision Support Systems (DSS) in Higher Education*, s.l.: s.n.
- Bader, S. e. a., 2020. A Knowledge Graph for Industry 4.0. In: C. B. T. B. S. Auer, ed. *The Semantic Web – ISWC 2020*. Cham: Springer: 465–480, p. 465–480.
- Chandak, P. H. K. & Z. M., 2022. Building a Knowledge Graph to Enable. *bioRxiv*, pp. 1–33.
- Galdamez, G., 2025. *Using Knowledge Management to Minimize the Costs of Departing Leaders*. [Online]
Available at: <https://enterprise-knowledge.com/using-knowledge-management-to-minimize-the-costs-of-departing-leaders/>
- Gao, Y. e. a., 2023. *Retrieval-Augmented Generation for LLMs: A Survey*, s.l.: arXiv.
- Gutiérrez, C. & S. J., 2021. Knowledge Grapghs. *Communications of the ACM*, 64(63), p. 96–104.
- Hak, P., Saddhono, K. & Pebriana, P., 2023. Integrating Decision Support Systems in Education: A Comprehensive Approach. *AL-ISHLAH: Jurnal Pendidikan*, 15(14), p. 4546–4552.
- Hamoud, A. K., Hussein, M. K., Alhilfi, Z. & Sabr, R. H., 2021. Implementing data-driven decision support system based on independent educational data mart. *International Journal of Electrical and Computer Engineering*, 11(6), p. 5301–5314.
- Lewis, D. D., Wang , Y. & Li,, x., 2016. A decision support system for predicting students’ performance. *International Journal of Advanced Computer Science and Applications*, 7(5), p. 190–196.
- Lewis, P. e. a., 2020. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. s.l., NeurIPS.

- Miron-Spektor, L. A. & E., 2011. Organizational Learning: From Experience to Knowledge. *Organization Science*, 22(5), pp. 1123-1137.
- Miron-Spektor, L. A. & E., 2011. Organizational Learning: From Experience to Knowledge. *Organizationa science*, 22(5), pp. 1123-1137.
- Neiroukh, S. E. O. & A. H., 2024. Artificial intelligence capability and organizational performance: unraveling the mediating mechanisms of decision-making processes. *Management Decision*, 62(10), p. 3501–3532.
- Saa, A. A., 2016. Educational data mining & students' performance prediction. *International Journal of Advanced Computer Science and Applications*, 7(5), pp. 212-220.
- Society, I., 2022. *Decision Support System to Help Educational Institutions*. Malaysia, IEOM Society.
- Systems, J. o. E., 2024. Design and Application of Decision Support System for Educational Management Based on Big Data. *Journal of Electrical Systems*, 20(2).
- Taherdoost, H. & M. M., 2023. AI and Knowledge Management: A Partnership. *Issues in Information Systems*, 24(4), p. 409–416.
- Wang, S. Y. H. & B. G., 2025. Construction of Intelligent Decision Support Systems through Integration of RAG and Knowledge Graphs. *Scientific Reports*, 15(15), p. 35462.
- Wikipedia, 2024. *Corporate amnesia*. [Online]
Available at: https://en.wikipedia.org/wiki/Corporate_amnesia
- Zhang, J. & Goyal, S. B., 2024. AI-Driven Decision Support System Innovations to Empower Higher Education Administration. *Journal of Computers, Mechanical and Management*, 3(4).
- Zhu, X. X. Y. L. Y. L. Y. & H. W., 2025. *Knowledge Graph-Guided Retrieval Augmented Generation*. s.l., NAACL-HLT 2025.
- Galan, N. (2023). Knowledge loss induced by organizational member turnover: A review of empirical literature, synthesis and future research directions (Part I). *The Learning Organization*, 117–136.
- Galan, N. (2023). Knowledge loss induced by organizational member turnover: A review of empirical literature, synthesis and future research directions (Part II). *The Learning Organization*, 137–161
- Panopto. (2018). *Workplace Knowledge and Productivity Report*. Panopto, Inc.

